KONGU ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JIRA SCRUM LAB

DETAILED REPORT OF EXPERIMENTS 1–10

Using the Alumnex Connect Scrum Project in Jira

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Institution: Kongu Engineering College, Perundurai

Project: Alumnex Connect

Jira project: ACS — Alumnex Connect - Scrum

Academic year: 2026–2027

Report snapshot date: 14 October 2026

Prepared from Jira issue exports, sprint/release metadata, repository source review and recorded test output.

Document Control

Field	Value
Report title	Detailed Report of Jira Scrum Experiments 1–10
Jira project	ACS — Alumnex Connect - Scrum (project ID 10033)
Jira board	ACS board (board ID 34)
Evidence snapshot	14 October 2026; current live state may change after the snapshot
Primary repository	https://github.com/SrimanKumarV/FSD/tree/master/alumni-connect-website-main
Live application entry point	https://alumnex-connect.onrender.com
Backend health endpoint configured in source	https://alumnex-backend-9y5t.onrender.com/api/health
Report scope	All ten faculty experiment topics, Jira evidence, source mapping, test results, limitations and export appendix
Evidence format	Structured Jira issue/JQL/sprint/version exports and embedded model diagrams; the report does not mislabel exported tables as browser screenshots

Executive Summary

This report documents the ten Jira Scrum experiments using the separate ACS project created for the Alumnex Connect lab. The report was prepared from current Jira issue records and JQL counts, sprint and release-version metadata, a study of the public GitHub repository, the deployment metadata available through Render, the recorded Python/MySQL lab test log, and the Alumnex-specific use-case/domain diagrams.

The Jira snapshot contains 46 issues: 4 Epics, 10 experiment Tasks, 15 product Stories and 17 Subtasks. At the snapshot, 19 issues are Done, 6 are In Progress and 21 are To Do. The ten experiment tasks are split between seven Done (Experiments 1–7) and three In Progress (Experiments 8–10). The master portfolio Epic, ACS-28, remains In Progress.

The product backlog now represents Alumnex Connect itself: a React frontend, Node.js/Express API, MongoDB/Mongoose persistence, Socket.IO/Redis real-time communication, AI career routes and a Capacitor Android wrapper. The 15 planned product Stories total 71 proposed points. These point values are educational planning estimates; they are not measured team velocity, and story status does not imply a feature has shipped.

> **Important scope distinction**
>
> The faculty Experiment 10 asks for a Python UI and MySQL database. The existing Alumnex Connect project is React / Node.js / Express / MongoDB. A separate Python/MySQL registration/login lab has been prepared to address the coursework requirement without falsely claiming that Python/MySQL is the production Alumnex stack.

The following external evidence items remain open: a successful real-MySQL run, a live interactive smoke test of the deployed demo, native saved JQL filters, and the final attachment of rendered UML files inside Jira. The report marks these clearly rather than asserting unverified completion.

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1. Introduction and Evidence Method

1.1 Objectives

- Perform and document the ten faculty-prescribed experiments using Jira Scrum capabilities and a separate ACS project.
- Build a product backlog with epics, user stories, acceptance criteria, priority and proposed story-point estimates grounded in Alumnex Connect modules.
- Configure releases, execute sprint lifecycle operations, record planning/review/retrospective evidence and show practical JQL query examples.
- Develop Alumnex-specific use-case and conceptual domain models based on repository source files and actual application modules.
- Document the separate Python/MySQL coursework lab and its actual test results, identifying the live database test that remains pending.

1.2 Source and evidence method

The Jira issue inventory and JQL results were read from the connected Atlassian site on 14 October 2026. Direct browse links are given for each experiment record. Project sprint data and release version data were read separately because issue search alone does not prove sprint or version lifecycle state. The repository was studied through the GitHub connector, including README, backend routes/models/services, frontend route map and workflow/test files. Render service metadata was used only as deployment-state evidence; it was not treated as proof of interactive functionality.

This report uses three kinds of evidence: (1) Jira configuration/record evidence, (2) source-code and test-file evidence, and (3) executed test output from the separate Python lab. The categories are not interchangeable. For example, the presence of a route in source is not proof that the corresponding external OAuth, push or calendar integration works against production services.

1.3 Status terminology

Term	Meaning in this report
Jira task Done	The Jira documentation/configuration exercise was recorded as completed. It does not imply every Alumnex feature is implemented or production-tested.
In Progress	The Jira experiment record is materially prepared but a stated evidence or practical step remains.
To Do	A planned or external verification step has not been performed.
Verified count	A count returned from a JQL query at the stated snapshot time; counts can change later.
Proposed story points	Planning estimates for the lab backlog; not completed effort or historical velocity.
Live / deployed	Deployment metadata reports a service/deploy live; this does not prove a successful browser-based user journey.

2. Project and Jira Environment

2.1 Jira project configuration

Property	Exported value
Site	https://srimankumar06.atlassian.net
Project name	Alumnex Connect - Scrum
Project key	ACS
Project ID	10033
Board	ACS board; board ID 34
Portfolio Epic	ACS-28 — Alumnex Connect — Scrum Lab Experiments 1–10
Total issues	46
Product Stories	15
Planned product Story points	71
Existing KAN project	Separate Kanban project; not used for this Scrum experiment set

2.2 Actual Alumnex Connect technology and capability areas

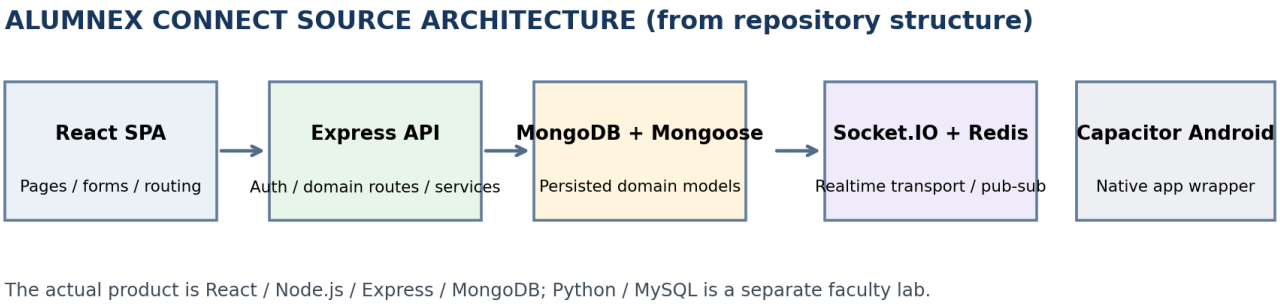


Figure 1. Source-derived architecture overview for Alumnex Connect. This is a report diagram based on repository structure, not a screenshot of the live running application.

The repository represents a multi-module alumni/student ecosystem rather than a simple student records application. The backend mounts route modules for authentication, users, mentorship, jobs, events, forum, contests, messages, admin, notifications, developer activity, leaderboards, projects, institutions, tasks, AI, Tech Hub, business, Activity Hub and Google Calendar. Mongoose models represent persisted entities; the React frontend routes correspond to user-facing pages; the backend has Jest/Supertest tests and the GitHub workflow defines backend tests, a frontend build, and Playwright smoke tests.

Capability	Representative source evidence	Backlog mapping
Identity & profile	backend/routes/auth.js, users.js; backend/models/User.js	ACS-1; ACS-12, ACS-13, ACS-37
Mentorship & career	backend/routes/mentorship.js; services/mentorshipService.js; routes/jobs.js	ACS-2; ACS-14, ACS-15, ACS-17, ACS-43
Developer intelligence	frontend/src/pages/DevPulse.js, ActivityHub.js; routes/devActivity.js, activity.js	ACS-3; ACS-16
Messaging & notifications	routes/messages.js, notifications.js; Socket.IO handler; Redis adapter setup	ACS-3; ACS-18
Events & calendar	routes/events.js, googleCalendar.js; GoogleCalendarConnection/Event models	ACS-3; ACS-19
Community / projects / contests	routes/forum.js, tech-hub.js, projects.js, contests.js, leaderboard.js	ACS-3; ACS-39, ACS-40, ACS-41
Mobile / push	Capacitor dependencies, .github/workflows/build-apk.yml, FCM service	ACS-3; ACS-42
Governance	routes/admin.js; auth middleware; incident/report models	ACS-1; ACS-38

2.3 Repository and deployment evidence

Repository root | README | Backend server | Frontend route map | CI workflow | Live demo | Backend health

Render service metadata reported the frontend and backend deployments as live from a commit dated 11 October 2026; GitHub check-runs for that deployment commit showed the Android build job completed successfully, while a separate Cloudflare Pages check failed. Direct interactive access to the live demo could not be retrieved from the report-generation runtime due to a DNS/cache fetch failure. Therefore, this report does not claim that every page or external integration was manually exercised.

3. Experiment 1 — Create a Product Backlog with Stories

Jira record: ACS-4 — Experiment 1: Product Backlog and Story Refinement. Status in the exported Jira snapshot: Done. The experiment created a source-grounded product backlog in project ACS and linked the ten-experiment portfolio under ACS-28.

3.1 Objective and approach

The goal was to define a backlog representing the working product domain shown in the repository. A generic course/enrolment/examination backlog would not match Alumnex Connect, so the backlog was reorganised into identity/profile/governance; mentorship/career; and developer-intelligence/community capabilities. Each Story is written from a user perspective and includes testable acceptance criteria, priority, a parent Epic, release grouping label and a proposed estimate.

3.2 Product Epics

Jira key	Epic	Purpose
ACS-1	Identity, Profiles and Governance	Registration, auth, profiles, connections, admin approval and moderation.
ACS-2	Mentorship and Career Opportunities	Mentorship lifecycle, job applications, AI career help, institution/business directory.
ACS-3	Developer Intelligence and Community Engagement	DevPulse/activity, messaging/notifications, events/calendar, forums, projects, contests and mobile.

3.3 Product user-story register

Story	Title	Parent	Release group	Priority	Points	Status
ACS-12	User Registration and Email Verification	ACS-1	Release 1.0	Medium	5	In Progress
ACS-13	Secure Login, OAuth and Session Lifecycle	ACS-1	Release 1.0	Medium	3	In Progress
ACS-37	Profiles, Connections and Role-Aware Access	ACS-1	Release 1.0	High	5	To Do
ACS-38	Admin Governance, Moderation and Analytics	ACS-1	Release 1.0	High	8	To Do
ACS-14	Mentorship Requests, Matching and Reviews	ACS-2	Release 1.1	Medium	5	To Do
ACS-15	Career Opportunities and Application Tracking	ACS-2	Release 1.1	Medium	3	To Do

Story	Title	Parent	Release group	Priority	Points	Status
ACS-17	AI Career Support and Resume Analysis	ACS-2	Release 1.1	Medium	5	To Do
ACS-43	Institution and Business Directory	ACS-2	Release 1.1	Medium	3	To Do
ACS-16	DevPulse, Developer Profiles and Activity Goals	ACS-3	Release 1.2	Medium	5	To Do
ACS-18	Real-Time Messaging and Notification Preferences	ACS-3	Release 1.2	Medium	3	To Do
ACS-19	Events and Google Calendar Synchronization	ACS-3	Release 1.2	Medium	3	To Do
ACS-39	Forum and Tech Hub Knowledge Sharing	ACS-3	Release 1.2	Medium	5	To Do
ACS-40	Project Showcase and Collaboration	ACS-3	Release 1.2	Medium	5	To Do
ACS-41	Competitive Contests and Leaderboards	ACS-3	Release 1.2	Medium	5	To Do
ACS-42	Capacitor Mobile App and Native Push Delivery	ACS-3	Release 1.2	High	8	To Do

The initial scope sums to 71 proposed points. Release 1.0 contains 21 points, Release 1.1 contains 16 points, and Release 1.2 contains 34 points. These are estimates for planning practice, not historical velocity or a statement that the Stories have been delivered.

3.4 Example user story and acceptance criteria

Example — ACS-12: User Registration and Email Verification

As a student/member, I need a validated registration and email-verification flow so that my account can be created and verified safely. Acceptance criteria include field validation; safe handling of duplicate/invalid account requests; email verification/resend flow; optional two-factor verification where enabled; and test evidence for both positive and negative paths.

Source mapping for the example: backend/routes/auth.js provides registration, email verification and verification-related routes; backend/services/authService.js contains authentication service logic; backend/models/User.js represents stored user data; and frontend/src/pages/auth/Register.js is the front-end registration entry. These links establish source traceability, while the actual end-to-end registration journey still requires live runtime testing.

3.5 Result and evidence

- Created three product Epics (ACS-1, ACS-2, ACS-3) and 15 product Stories.
- Each Story has a user outcome, acceptance criteria, priority and proposed Story Point estimate.
- Parent relationships assign Stories to the relevant product Epic.
- Release labels group stories into three proposed releases because standard Fix Version assignment was not available through the connected editing interface.

- Current status is still a backlog snapshot: only ACS-12 and ACS-13 are In Progress; the remaining 13 product Stories are To Do.

Experiment 1 conclusion: The backlog design and Jira records are complete. The Done status applies to backlog preparation, not to implementation of all 15 product Stories.

4. Experiment 2 — Determine the Release Plan

Jira record: ACS-5 — Experiment 2: Release Plan. Status: Done. The release plan was based on dependencies and user-facing value in the actual Alumnex Connect product: identity and access first, career workflows next, then developer/community engagement.

4.1 Release plan

Planned version / ID	Focus	Stories	Points	Jira state
Release 1.0 — Identity and Profile Foundations (10037)	Registration, email verification, login/OAuth/session lifecycle, profile/connections and admin governance.	ACS-12, 13, 37, 38	21	Unreleased
Release 1.1 — Mentorship and Career Enablement (10038)	Mentorship requests/matching/reviews, jobs/applications, AI career help, institution/business directory.	ACS-14, 15, 17, 43	16	Unreleased
Release 1.2 — Developer Intelligence and Community (10039)	DevPulse/activity, messaging/notifications, events/calendar, forum/Tech Hub, projects, contests, mobile/push.	ACS-16, 18, 19, 39–42	34	Unreleased
Total planned backlog		15 Stories	71	Not a commitment

4.2 How the releases were decided

1. Release 1.0 focuses on account access and ownership because profile, mentorship, career and activity features depend on authenticated member identities and role permissions.
2. Release 1.1 expands value through mentorship and employment workflows. These capabilities depend on users/profiles and require service-level validation of matching/application state and appropriate authorization.
3. Release 1.2 brings together activity and developer data, notifications, community content, events/calendar and the Android wrapper. This scope has more external/provider dependencies and should be released only after integration and device checks.

4.3 Jira version configuration and limitation

Jira project ACS contains version records with IDs 10037, 10038 and 10039. Their names/descriptions were updated and all remain unreleased. The story groups use explicit labels release-1.0, release-1.1 and release-1.2. The team-managed project metadata available through the connected operation did not permit standard Fix Version updates, so the labels are used as grouping—not represented as native issue Fix Version assignments.

4.4 Evidence

- ACS-5 stores the release strategy and traceability links to Stories.
- Jira release metadata confirms three unreleased versions.
- ACS-8/ACS-30 queries return 4, 4 and 7 Stories for release groups 1.0, 1.1 and 1.2, respectively.
- Release dates have not been committed and no version is marked released.

5. Experiment 3 — Write the Sprint Plan

Jira record: ACS-6 — Experiment 3: Sprint Planning for Alumnex Connect. Status: Done. The Scrum board is ACS board (ID 34). The record contains the historical Sprint 1 setup and the active Sprint 2 plan.

5.1 Sprint 1 — Accounts

Planning field	Jira export / recorded value
Sprint ID / name	1 — Sprint 1 - Accounts
State	Closed
Goal	Deliver an initial student-account increment by preparing and validating registration and login.
Selected Stories	ACS-12 Registration (5 points); ACS-13 Login (3 points)
Planned load	8 proposed points (course-lab capacity assumption, not historical velocity)
Dates planned	13–20 October 2026
Actual lifecycle	Started and closed on 13 October 2026; 0 of 2 Stories accepted
Interpretation	Lifecycle exercise completed; the intended product increment was not accepted.

5.2 Sprint 2 — Alumnex Core

Planning field	Jira export / recorded value
Sprint ID / name	2 — Sprint 2 — Alumnex Core
State	Active
Start / end	13 October 2026 to 20 October 2026; Jira timestamps correspond to 09:30–18:00 IST
Goal	Deliver a testable vertical slice across registration/email verification, login/profile access, mentorship requests and career opportunity/application handling.
Scope	ACS-12 (5), ACS-13 (3), ACS-14 (5), ACS-15 (3)
Planned load	16 points; proposed planning load, not measured velocity
Current statuses	2 In Progress (ACS-12, ACS-13); 2 To Do (ACS-14, ACS-15); 0 Done
Assignment	Four Stories assigned to SRIMAN VEL in the current snapshot

5.3 Planning method

- Choose a bounded sprint goal that can be demonstrated and tested, rather than trying to finish every module in the portfolio.
- Use existing acceptance criteria and define dependencies before starting work.
- Break work into implementation/testing tasks without assuming subtasks imply feature acceptance.
- Track the sprint goal alongside per-Story status and story-point estimates.
- Use the Definition of Ready and Definition of Done recorded in ACS-33 as the proposed lab guardrails.

6. Experiment 4 — Manage Workload by Executing the Sprint Plan

Jira record: ACS-7 — Experiment 4: Sprint Execution and Workload Management. Status: Done as the Jira execution/documentation exercise. The record documents the sprint lifecycle, assignments, breakdown and the distinction between planned work and verified product acceptance.

6.1 Execution controls recorded

Control	Current evidence / meaning
Sprint scope	Active Sprint 2 contains ACS-12 to ACS-15; 16 proposed points.
Work states	ACS-12 and ACS-13 are In Progress; ACS-14 and ACS-15 are To Do; no active-sprint Story is Done.
Task decomposition	ACS-22–ACS-27 split registration/login work into validation, endpoint contract, credential verification, session/token review and test planning/cases; ACS-34 records offline lab tests; ACS-35 tracks live MySQL.
Workload ownership	The four Sprint 2 Stories are assigned; related technical tasks are visible as subtasks under their parent work.
Acceptance integrity	Sprint 1 ended with 0/2 Stories accepted. Stories were not falsely set Done without acceptance evidence.
Product source/test grounding	Repository contains auth/service/role tests, mentorship service tests, job service tests, Activity Hub reliability tests, Google Calendar tests, mobile OAuth and FCM token handling tests.

6.2 Sprint status snapshot

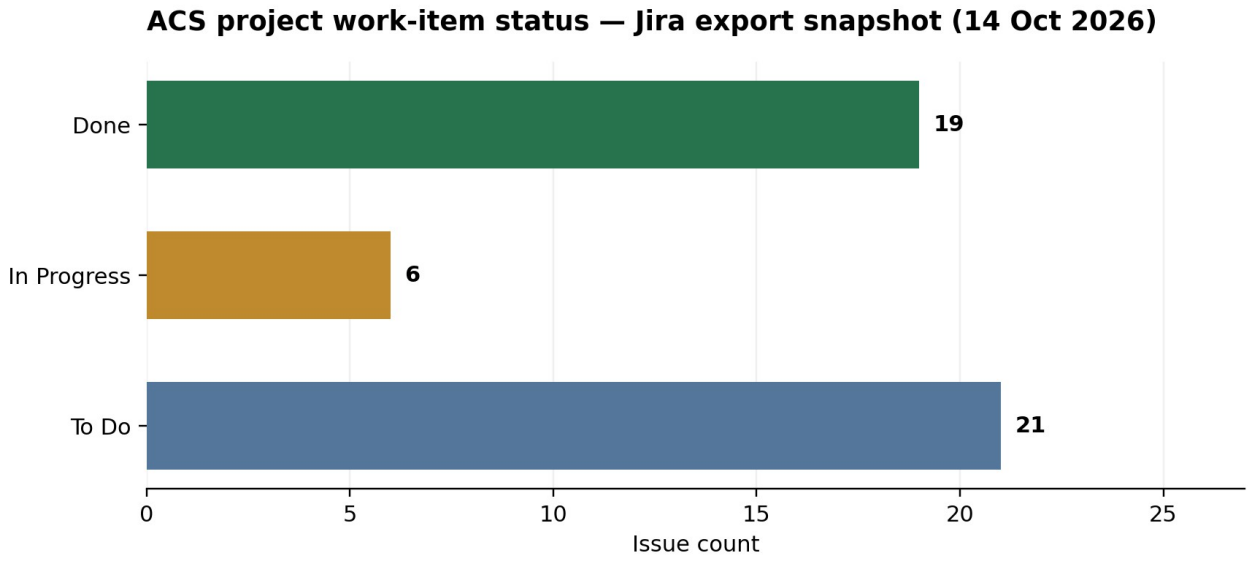


Figure 2. Work-item status distribution computed from the Jira issue export at the report snapshot. This is a report-generated chart of exported fields, not a screenshot of Jira's board UI.

The status distribution is a project-wide snapshot, not a sprint burndown chart. It should not be interpreted as delivery velocity or percent-complete for Alumnex features. The 46 issues include experiment records and administrative/follow-up subtasks as well as product backlog Stories.

6.3 Workload operating procedure

4. At the start of each working day, inspect the active sprint and each assigned Story's acceptance criteria.
5. Each contributor reports work completed, next action and blocker. Update status only when the related work state has changed.
6. When coding work is finished, run the appropriate tests. Distinguish unit tests, offline integration tests, database integration tests, external-provider checks and device tests.
7. During review, compare the output to the acceptance criteria and DoD; move only accepted Stories to Done.
8. For blocked items, record the dependency as a child issue and preserve the unresolved state until the dependency is verified.

6.4 Result

The sprint was operated using real Jira status fields, assignments, estimates and child tasks. The recorded issue completion state is honest: the workflow/documentation experiment is Done, while the product increment is not represented as shipped or accepted. The source repository README reports 56 backend tests, but the report does not present that count as a fresh test run; it is a claim from the README.

7. Experiment 5 — Use Predefined and User-Created Queries

Jira record: ACS-8 — Experiment 5: JQL Queries. Status: Done for query execution/documentation. ACS-29, ACS-30 and ACS-31 divide the examples into sprint/status, release/experiment, and backlog/defect queries. Five links in the issue Web links section open direct JQL searches.

7.1 Predefined and custom JQL set

Purpose	JQL query	Exported count
All ACS project issues	project = ACS	46
Product backlog stories	project = ACS AND issuetype = Story	15
Active sprint scope	project = ACS AND Sprint in openSprints()	4
Active sprint incomplete	project = ACS AND Sprint in openSprints() AND statusCategory != Done	4
Active sprint completed	project = ACS AND Sprint in openSprints() AND statusCategory = Done	0
Unresolved bugs	project = ACS AND issuetype = Bug AND statusCategory != Done	0
Release 1.0 label group	project = ACS AND labels = "release-1.0"	4
Release 1.1 label group	project = ACS AND labels = "release-1.1"	4
Release 1.2 label group	project = ACS AND labels = "release-1.2"	7
Experiment 8 label	project = ACS AND labels = "experiment-8"	1
Experiment 9 label	project = ACS AND labels = "experiment-9"	1
Experiment 10 label	project = ACS AND labels = "experiment-10"	1

7.2 How to interpret the counts

- project = ACS returns 46 issues as of the snapshot, including experiment tasks, product Stories, Epics and subtasks.
- project = ACS AND issuetype = Story returns 15 user Stories, the product backlog used in release planning.
- Sprint in openSprints() returns four active Sprint 2 Stories. All four are incomplete at the snapshot.
- The release labels return 4, 4 and 7 Stories. These are grouping labels because native Fix Version assignment was not available via the connected edit tool.
- The unresolved Bug query returns zero issues. This means there are no matching Jira Bug issues in this project snapshot; it does not prove there are no defects in the product.

7.3 Predefined query links and user-created saved filters

ACS-8 includes direct query launchers named Product Stories, Active Sprint, Active Sprint Incomplete, Release 1.0 Stories and Unresolved Bugs. These links open searches with the stated JQL. They are not the same as Jira's native saved filters. The connected operation set did not expose a filter-create operation, so the native "Save as" action remains an external UI step tracked in ACS-44.

Reproduction instructions

Open Jira Issue Search, paste a query from the table, run it, and choose "Save as" to make a reusable native filter. Verify the filter owner and sharing level. Do not describe a query link as a saved filter until the filter exists in Jira.

8. Experiment 6 — Prepare a Schedule for Reviewing the Sprint

Jira record: ACS-9 — Experiment 6: Sprint Review. Status: Done for agenda and review-planning records. Child issue ACS-36 has the review agenda and due date set to 20 October 2026.

8.1 Review setup

Field	Recorded plan
Review topic	Sprint 1 — Accounts outcome and Sprint 2 work
Planned due date	20 October 2026
Suggested time	16:30–17:00 IST, subject to confirmation
Jira child record	ACS-36 — Sprint Review — Agenda and Date
Stakeholders	Faculty/Product Owner, Scrum team, demonstrator/reviewer
Invitation state	No calendar invitation or attendance was established by the Jira operation; time is proposed only.
Sprint 1 result to present	0 of 2 Stories accepted; no verified product increment was available at closure.

8.2 Proposed 30-minute agenda

Time	Activity	Expected outcome
0–5 min	Restate sprint goal and selected stories	Common scope and acceptance criteria.
5–15 min	Demonstrate only currently working Alumnex flows	Evidence of actual behavior; no mocked-up feature presented as live.
15–20 min	Review acceptance criteria against tests/results	Accepted/not-accepted decisions.
20–25 min	Collect faculty/Product Owner feedback	New backlog items and priority changes.
25–30 min	Record outcome and next actions	Decision log and follow-up task owners.

8.3 Review checklist

- Authentication: registration, email verification, login, OAuth callback and role-gated routes.
- Mentorship: mentor discovery/request and state transitions with safe test accounts.
- Career: job discovery/application and invalid/duplicate response handling.
- DevPulse, activity, chat, notifications and Google Calendar: use actual database and provider responses where available.
- Cross-cutting behavior: owner isolation, user-safe errors, empty/loading states, and desktop/mobile rendering.

Experiment 6 conclusion

The review agenda and proposed date are recorded. The meeting itself is not claimed to have occurred, and the 16:30–17:00 time must be confirmed by faculty/team.

9. Experiment 7 — Close Sprint 1 and Start the Next Sprint

Jira record: ACS-10 — Experiment 7: Sprint Closure and Next Sprint. Status: Done. The native Jira Scrum lifecycle was used to close Sprint 1 and start Sprint 2; the retrospective and DoR/DoD records are retained as related work.

9.1 Sprint lifecycle record

Lifecycle attribute	Sprint 1 — Accounts	Sprint 2 — Alumnex Core
Sprint ID	1	2
Jira state	Closed	Active
Goal	Initial student-account registration/login increment	Testable Alumnex vertical slice across registration/login, mentorship and career application workflows
Planned points	8	16
Selected Stories	ACS-12, ACS-13	ACS-12, ACS-13, ACS-14, ACS-15
Actual completion snapshot	0 of 2 Stories accepted	0 of 4 Stories Done; 2 In Progress and 2 To Do
Dates	13–20 Oct planned; closed 13 Oct	13–20 Oct planned; active at snapshot

9.2 Retrospective and improvements

ACS-32 records the Sprint 1 retrospective. The central observation is that the sprint lifecycle was demonstrated, but the product increment was not accepted. Improvements include aligning the lab backlog to real Alumnex modules, preparing integration setup earlier, recording test output before review, tracking OAuth/provider/device dependencies, and separating the production stack from the faculty Python/MySQL lab.

ACS-33 documents proposed Definition of Ready and Definition of Done guardrails. A Story should be Ready only when its outcome and acceptance criteria are clear, its parent/priority/estimate/dependencies are documented, and the source/test path is identified. A Story should be Done only when acceptance criteria are satisfied and test evidence exists for the feature claimed, including real external dependencies where required.

9.3 Interpretation

This experiment shows a genuine Jira lifecycle transition. Closing a sprint is not the same as accepting all scope. The honest result of Sprint 1 was 0/2 Stories accepted, and Sprint 2 remains active with four planned Stories incomplete.

10. Experiment 8 — Identify Use Cases and Develop a Business Use-Case Model

Jira record: ACS-11 — Experiment 8: Alumnex Connect Business Use-Case Model. Status: In Progress. The textual and Mermaid source model is in Jira. The rendered diagram file is in the companion artifact package; attaching the binary directly to Jira is tracked in ACS-46.

10.1 System boundary

The use-case model treats Alumnex Connect as the system boundary for the React web/mobile client and Express backend. MongoDB/Mongoose is an internal persistence component, not a human actor. External identity/email/calendar/developer/AI/push providers are shown only where source modules describe integrations; inclusion in the model is not a claim that credentials or production connectivity were verified.

10.2 Actors

Actor	Primary interactions with the system
Student / Member	Register/sign in, maintain profile, find mentors, request mentoring, search/apply to jobs, use AI career tools, view DevPulse/activity, communicate and use events/community functions.
Alumni / Mentor	Maintain professional profile and availability, respond to mentorship requests, manage sessions/milestones, review mentees and manage job opportunities where authorized.
Administrator	Approve users/mentors, manage roles/status, moderate content, inspect analytics/system incidents and send admin notifications.
Institution / Business	Participate in the directory/institutional and business capabilities where configured.
Google / GitHub OAuth	External identity provider for supported OAuth sign-in paths.
Email / OTP provider	Supports verification, account recovery and optional 2FA delivery flows.
Developer APIs	External sources for developer profiles/activity (e.g. GitHub, LeetCode, Codeforces where supported).
AI model providers	Groq / Gemini provider paths for AI career assistance when configured.
Google Calendar	Connect/list calendars, read/write/sync events and provide schedule context.
Web Push / FCM	Browser and Android notification delivery boundaries.

10.3 Use-case inventory

- Identity and access: registration, login, OAuth completion, role selection, email verification, password recovery/change, optional 2FA, refresh and logout.
- Member network: search/view profiles, update own profile, connect/follow, accept/decline requests and manage status/presence.
- Mentorship: discover mentor, request/auto-assign mentor, change status, schedule session, manage milestones/notes, cancel and review.

- Career: browse internal/external openings, save/apply to a job, manage application status and use resume analysis/job match/mock-interview features.
- Developer momentum: manage developer profile, sync activity, goals/preferences, view timeline/streaks/analytics/insights and export/delete own data.
- Communication: start conversations, send/read/search messages, manage groups and read/subscribe/unsubscribe/register notifications/devices.
- Events/calendar: register for events, manage event state where authorized, connect Google Calendar, manage settings/calendars/events, sync and obtain schedule context.
- Community and portfolio: forum feed/search/comments/reports, Tech Hub questions/replies, project showcase, contests/leaderboards and directory search.
- Governance: approve and verify users, assign roles, suspend accounts, moderate content, view admin analytics and investigate incidents.

10.4 Use-case model figure

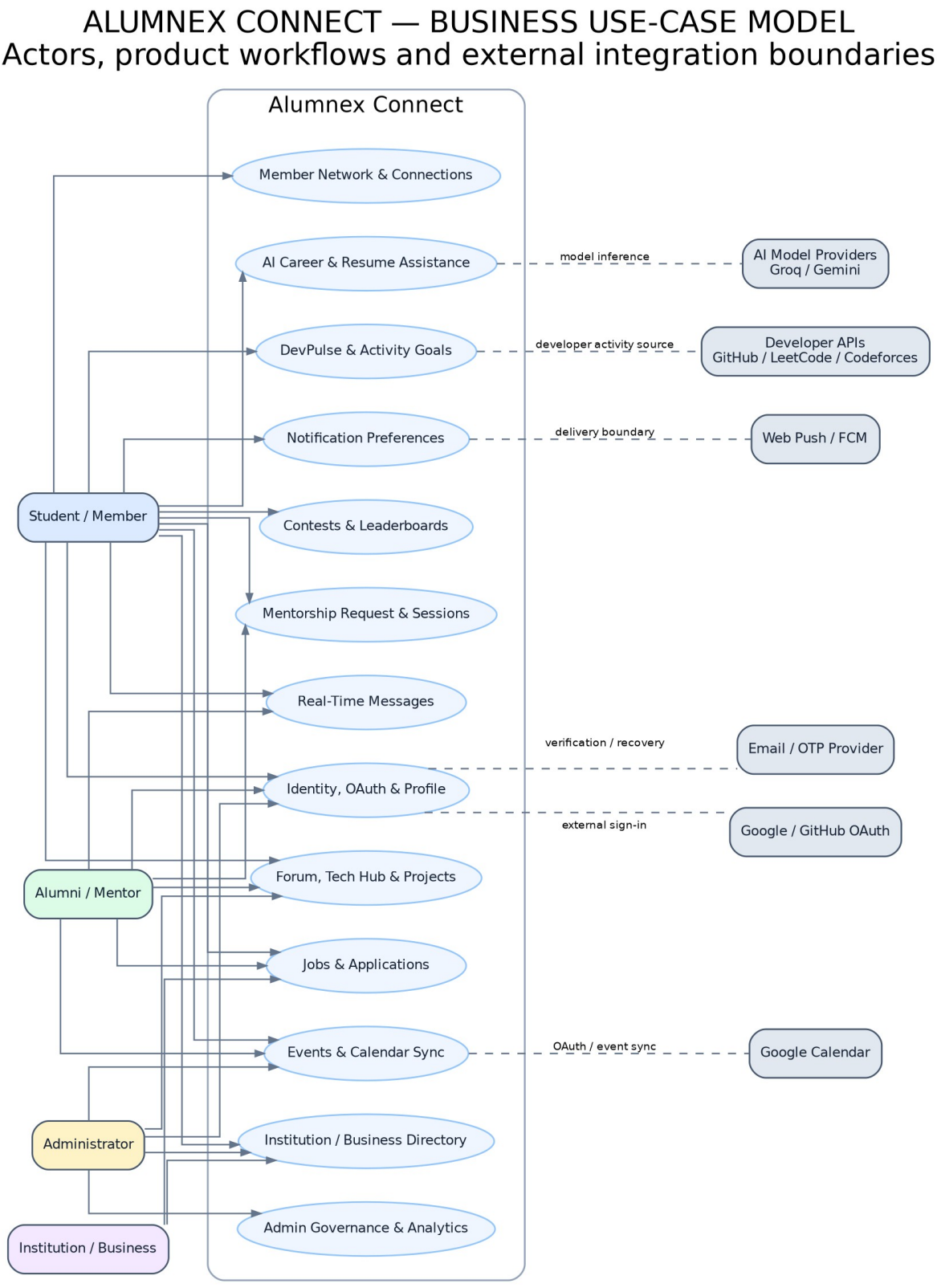


Figure 3. Alumnex Connect business use-case model. Rendered from the companion package's source-derived diagram; its binary has not yet been attached to Jira.

10.5 Relationships and controls

Authentication and authorization act as preconditions for restricted actions. Mentorship requests interact with mentorship/session/review entities and may trigger notifications. Job applications link a user to a job. Developer activity sync writes activity records and goal-related state. Messaging uses persisted records plus Socket.IO transport; Redis adapter configuration supports cross-instance pub/sub. Calendar workflows use Google OAuth and user-specific connection/event state. AI workflows call external providers and must handle provider failures safely.

Experiment 8 conclusion

The actors, system boundary, use-case inventory and Mermaid source have been documented in Jira. The experiment is kept In Progress until the diagram export is attached and the visual evidence is available directly on the Jira issue.

11. Experiment 9 — Conceptual Classes, Boundary/Control/Entity and UML Class Diagram

Jira record: ACS-20 — Experiment 9: Alumnex Connect Conceptual Classes and UML Domain Model. Status: In Progress. The model is grounded in the actual Mongoose model folder, React page routes and Express route/service modules.

11.1 Boundary classes

Boundary roles represent the interaction surfaces already present in the React application; they are conceptual role labels and not necessarily separate class declarations. Representative surfaces include Login/Register, Profile/Network, Mentorship, Jobs/Resume Analyzer, DevPulse/Activity Hub, Chat/Notifications, Events/Calendar, Forum/Tech Hub/Projects, Contests and Admin.

11.2 Control classes/modules

The Control layer uses existing Express route modules and services rather than inventing a separate MVC controller class for every page. Authentication/users, mentorship, jobs/AI, activity/developer data, messages/notifications, events/Google Calendar, forum/Tech Hub/projects/contests/institutions/business and admin routes coordinate requests, validation, authorization, business logic and persistence.

11.3 Entity classes

Domain group	Representative entities	Key responsibilities
Identity / mentorship	User, MentorshipRequest, Mentorship, MentorshipSession, MentorReview, MentorReward	Identity/role/profile, request lifecycle, sessions and feedback.
Career / applications	Job, JobApplication, JobOpportunity	Job details, applications, status and applicant/job association.
Developer / activity	DevProfile, ActivityGoal, ActivityRecord	External platform identity, goals, activity history and trust/source data.
Messaging / notifications	Message, ChatGroup, Notification, NotificationDevice, ReminderPreference	Conversations, recipient-scoped notices, device and delivery preferences.
Events / calendar	Event, GoogleCalendarConnection, GoogleCalendarEvent	Event lifecycle and per-user Google Calendar connection/sync records.
Community / governance	ForumPost, Project, Contest, Business, TechHub entities, Report, AdminIncident	Member content, portfolio links, contest workflows, moderation and incidents.

11.4 Conceptual relationships

- One User can act as student and/or mentor across multiple mentorship requests/relationships.
- A Mentorship can have sessions and feedback records; reward totals are associated with mentor outcomes.
- One Job can have multiple JobApplication records; each application is associated with one applicant and one job.
- A User owns developer profile/activity/goals and notification preferences/device registrations.
- Messages have sender/recipient context and may belong to a conversation/group; notifications have a recipient and may refer to a related event/action.
- A user-specific GoogleCalendarConnection relates a member to a calendar provider; GoogleCalendarEvent stores external event identity/sync context.
- User-owned ForumPost and Project records use author/owner references to establish edit/delete boundaries.

11.5 UML domain model figure

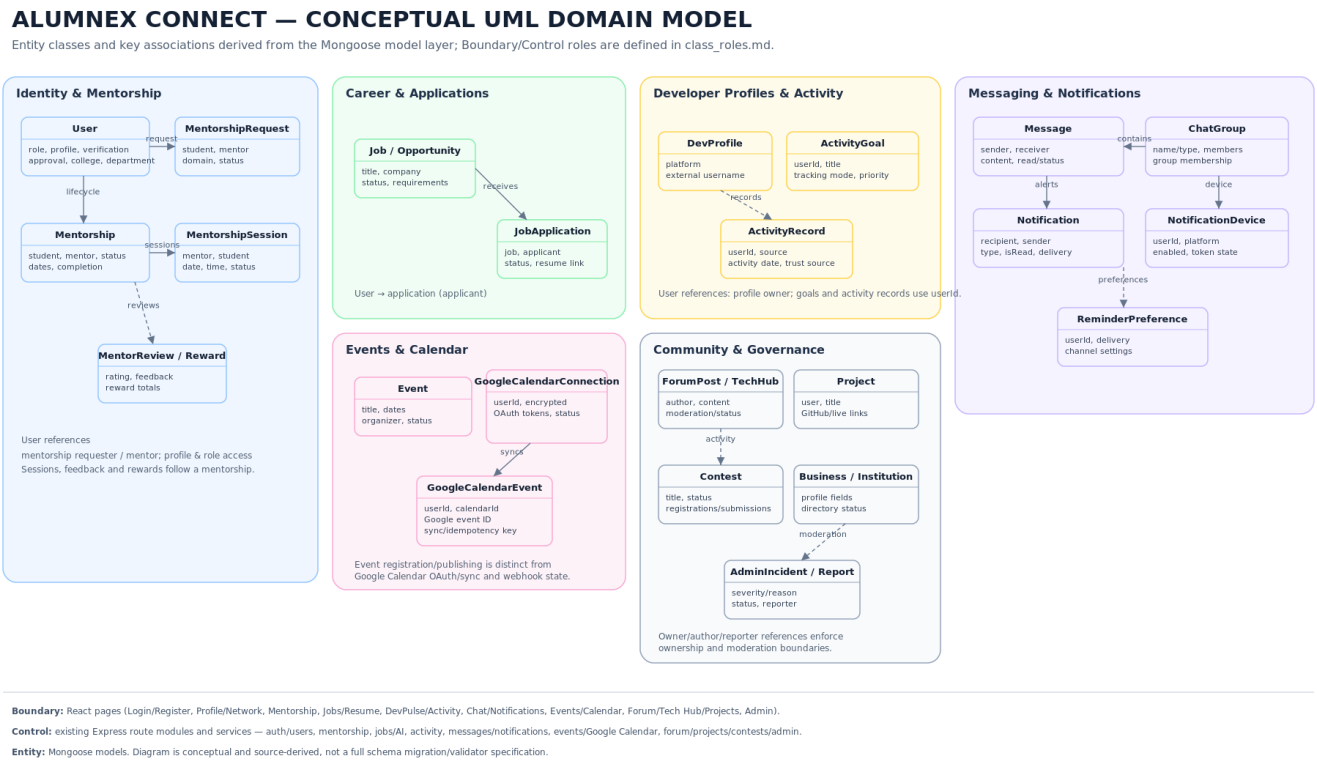


Figure 4. Alumnex Connect conceptual UML domain model. This is a source-derived conceptual model, not an exact export of every Mongoose validator/index or a database migration diagram.

11.6 Limitations

The model provides a useful conceptual view of entity groups and associations; it does not replace inspection of full Mongoose schema definitions for constraints, indexes, reference policies or optional fields. The diagram export remains pending attachment to ACS-20; ACS-46 tracks both Experiment 8 and 9 diagram attachments.

Experiment 9 conclusion

The Boundary/Control/Entity classification and UML source have been prepared from Alumnex Connect. The Jira task remains In Progress until the rendered UML file is attached for easy in-issue inspection.

12. Experiment 10 — Python UI, MySQL Database, Unit and Integration Testing

Jira record: ACS-21 — Experiment 10: Alumnex Source Study, Python/MySQL Lab and Testing. Status: In Progress. This experiment has two distinct contexts that must be kept separate: the actual Alumnex Connect production codebase and the Python/MySQL coursework lab built for the faculty's stated requirement.

12.1 Actual Alumnex Connect stack

Layer	Repository evidence / technology
Web UI	React 18, React Router, React Query, TailwindCSS, Framer Motion and Recharts.
API / backend	Node.js and Express route modules/services.
Persistence	MongoDB Atlas or configured MongoDB URI through Mongoose models.
Realtime	Socket.IO and Redis adapter/pub-sub configuration.
AI / external APIs	Groq/Gemini provider paths and developer-platform integrations where configured.
Mobile	Capacitor Android wrapper and FCM/notification integration.
Testing / CI	Backend Jest/Supertest tests; frontend build; Playwright smoke workflow defined in .github/workflows/test.yml.

The actual project does not use Python/MySQL as its production UI/database stack. Therefore the Python/MySQL sample is presented as a separate educational lab, not as a claim about Alumnex's production implementation.

12.2 Python/MySQL coursework lab structure

File / component	Role
python_mysql_lab/app.py	Python WSGI application with HTML home/register/login UI, JSON health/register/login routes, request parsing and database injection seam.
python_mysql_lab/validation.py	Required-field, email-normalization and password validation.
python_mysql_lab/security.py	PBKDF2-HMAC-SHA256 password hashing and verification helpers.
python_mysql_lab/schema.sql	Creates the dedicated acs_student_lab database and users table with unique email and password_hash columns.
tests/test_unit.py	Unit tests for validation, email normalization, hash verification and malformed hash handling.
tests/test_wsgi_integration.py	Offline request-to-data-access tests using an in-memory fake database.
tests/test_mysql_integration.py	Opt-in live MySQL registration/login test; skipped unless RUN_MYSQL_INTEGRATION=1 and connection settings are provided.
setup_xampp.md	Windows/XAMPP steps for configuring MySQL and recording the successful test output.

12.3 Database design

The sample schema is deliberately small and isolated from any production database. It uses the database name acs_student_lab and table users. The primary key is an auto-incrementing BIGINT; full_name is required; email is unique; password_hash stores a hash rather than a plain-text password; created_at records creation time.

```
CREATE DATABASE IF NOT EXISTS acs_student_lab
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE acs_student_lab;
CREATE TABLE IF NOT EXISTS users (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(120) NOT NULL,
  email VARCHAR(254) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

12.4 Security and validation design

- Use parameterized SQL rather than interpolating user input into SQL statements.
- Normalize email before validation and persistence; reject missing name, invalid email, and passwords outside allowed length rules.
- Reject duplicate emails through both application checks and a database UNIQUE constraint.
- Use PBKDF2-HMAC-SHA256 password hashes; never save or render the plain-text password.
- Return generic login failures and user-safe server errors; do not return raw database exception details.
- Use environment variables for connection configuration and run only against a dedicated educational database.

12.5 Actual test run and results

Test category	Result	Coverage
Unit tests	8 passed	Email normalization, valid/invalid email, missing name, password limits, hash verification and malformed hash.
Offline WSGI integration	7 passed	Home/health routes, registration/duplicate handling, invalid payload before DB access, successful/failed login, unknown user, simulated DB outage, HTML escaping.
Live MySQL integration	1 skipped	Real registration/login persistence test was not run because a reachable/configured MySQL server and connector were unavailable in the execution environment.
Total	16 discovered; 15 passed; 1 skipped; 0 failures	Recorded in test-results.txt on 14 October 2026.

Actual test command

Run from python_mysql_lab: PYTHONPATH=. python -m unittest discover -s tests -v

The offline suite is successful within its scope, but the skipped test means real MySQL connectivity and persistence remain unverified. A fake database validates request flow and SQL contract behavior; it does not substitute for the actual MySQL server integration criterion.

12.6 Steps to complete the real MySQL integration

1. On Windows, start MySQL from XAMPP Control Panel.
2. Open phpMyAdmin and execute schema.sql to create the separate acs_student_lab database/table.
3. Install requirements with python -m pip install -r requirements.txt.
4. Set DB_HOST, DB_PORT, DB_NAME=acs_student_lab, DB_USER and DB_PASSWORD in the test PowerShell session. Use a local least-privilege educational account; do not use the production database.
5. Set RUN_MYSQL_INTEGRATION=1 and PYTHONPATH=. then run python -m unittest tests.test_mysql_integration -v.
6. Save the real console output and attach/comment it in ACS-35. Mark ACS-35 and ACS-21 Done only if the live test passes.

12.7 Acceptance checklist

Requirement	Status at snapshot	Evidence
Alumnex source stack researched	Complete	Repository routes/models/README and architecture notes linked in ACS-21 and ACS-28.
Python user interface prepared	Complete for separate lab	python_mysql_lab/app.py.
MySQL schema prepared	Complete; not live-verified	python_mysql_lab/schema.sql.
Unit tests executed	Complete	8 unit test cases passed within 16-test suite.
Offline WSGI integration tests executed	Complete	7 route/SQL-contract tests passed.
Real MySQL integration test	Pending	ACS-35, To Do; one test skipped in latest run.
Live Alumnex UI smoke test	Pending	ACS-45, To Do; runtime could not load the demo host.
Experiment 10 overall	In Progress	Do not claim full completion until live MySQL test passes.

Experiment 10 conclusion

The code, schema, unit tests and offline integration tests are prepared and recorded. A full completion claim is not supported until the test runs successfully against a real MySQL instance and its output is attached to ACS-35.

13. Overall Results and Remaining Work

13.1 Experiment completion snapshot

No.	Faculty topic	Issue	Status	Summary of evidence
1	Product backlog with stories	('ACS-4', 'https://srimankumar06.atlassian.net/browse/ACS-4')	Done	Backlog includes 3 product Epics and 15 Stories, acceptance criteria, priority, proposed estimates and release groups.
2	Release plan	('ACS-5', 'https://srimankumar06.atlassian.net/browse/ACS-5')	Done	3 unreleased Jira versions; stories grouped by release labels; 71 proposed points total.
3	Sprint planning	('ACS-6', 'https://srimankumar06.atlassian.net/browse/ACS-6')	Done	Sprint 1 planned; Sprint 2 active with a goal, scope and 16 proposed points.
4	Execute sprint/manage workload	('ACS-7', 'https://srimankumar06.atlassian.net/browse/ACS-7')	Done	Sprint lifecycle/workload documentation and task breakdown recorded; Sprint 1 accepted 0/2 Stories.
5	Queries to track progress	('ACS-8', 'https://srimankumar06.atlassian.net/browse/ACS-8')	Done for execution/docs	JQL results and five query launch links; native saved filters still a manual UI action.
6	Schedule sprint review	('ACS-9', 'https://srimankumar06.atlassian.net/browse/ACS-9')	Done for agenda	Agenda and proposed 20 Oct date stored; no calendar invitation/attendance claimed.
7	Close first/start next sprint	('ACS-10', 'https://srimankumar06.atlassian.net/browse/ACS-10')	Done	Sprint 1 closed, Sprint 2 active; retrospective and DoR/DoD records.
8	Use cases and model	('ACS-11', 'https://srimankumar06.atlassian.net/browse/ACS-11')	In Progress	Alumnex-specific model source in Jira; binary image attachment pending.
9	Conceptual classes and UML	('ACS-20', 'https://srimankumar06.atlassian.net/browse/ACS-20')	In Progress	Source-derived model and Boundary/Control/Entity notes; binary attachment pending.
10	Python UI + MySQL + testing	('ACS-21', 'https://srimankumar06.atlassian.net/browse/ACS-21')	In Progress	Offline tests pass; one live MySQL test skipped and demo UI runtime check pending.

13.2 Outstanding practical items

Follow-up	Jira key	Current state	Required close-out evidence
Save native JQL filters	ACS-44	To Do	Save five JQL searches as native filters in Jira UI; record filter names/URLs and sharing settings.

Follow-up	Jira key	Current state	Required close-out evidence
Live demo smoke test	ACS-45	To Do	Use authorized test account and record results per route/flow, with screenshots or safe console/network excerpt.
Attach UML exports	ACS-46	To Do	Attach use_case_model.png/svg to ACS-11 and domain_model.png/svg to ACS-20; verify files show in Jira attachments.
Run actual MySQL test	ACS-35	To Do	Run live integration against local dedicated MySQL/XAMPP schema, save successful console output and attach/comment it.
Confirm review meeting	ACS-36	Due date set; meeting not confirmed	Confirm time with faculty/team and record actual attendance/outcome after meeting.

13.3 Conclusion

The Jira Scrum experiment set is organised around an authentic Alumnex Connect product model, with three product Epics, 15 product Stories, staged release groups, two Sprint lifecycle records, JQL evidence and source-derived use-case/domain models. The report preserves the actual exported status snapshot and distinguishes the completed Jira documentation exercises from unverified product or external-integration behavior. The remaining To Do tasks are specifically identified so that practical confirmation can be completed and documented before the report is treated as fully closed.

Appendix A — Jira Export Summary

The tables below are exported from Jira reads on 14 October 2026. They are report-formatted structured extracts, not direct browser screenshots. Issue keys link to their live Jira browse pages.

JIRA EXPORT SNAPSHOT | ACS EXPERIMENT TRACKER

Rendered from exported issue fields; not a direct browser screenshot. Snapshot date: 14 Oct 2026.

KEY	ISSUE SUMMARY	TYPE	STATUS	PRIORITY
ACS-4	Experiment 1 — Product Backlog and Story Refinement	Task	Done	High
ACS-5	Experiment 2 — Release Plan	Task	Done	High
ACS-6	Experiment 3 — Sprint Planning for Alumnex Connect	Task	Done	High
ACS-7	Experiment 4 — Sprint Execution and Workload Management	Task	Done	High
ACS-8	Experiment 5 — JQL Queries	Task	Done	High
ACS-9	Experiment 6 — Sprint Review	Task	Done	High
ACS-10	Experiment 7 — Sprint Closure and Next Sprint	Task	Done	High
ACS-11	Experiment 8 — Alumnex Connect Business Use-Case Model	Task	In Progress	High
ACS-20	Experiment 9 — Alumnex Connect Conceptual Classes and UML Domain Mod..	Task	In Progress	High
ACS-21	Experiment 10 — Alumnex Source Study, Python/MySQL Lab and Testing	Task	In Progress	High
ACS-28	Alumnex Connect — Scrum Lab Experiments 1–10 Portfolio	Epic	In Progress	High
ACS-35	Experiment 10 — Run Against Real MySQL	Subtask	To Do	High
ACS-44	Experiment 5 — Save Native JQL Filters in Jira	Subtask	To Do	High
ACS-45	Experiment 6 — Run Live Alumnex Demo Smoke Test	Subtask	To Do	High
ACS-46	Experiments 8–9 — Attach UML Diagram Exports to Jira	Subtask	To Do	High

Data fields: issue key, summary, type, status, priority. Full CSV and JSON exports are provided separately.

Figure 5. Selected Jira experiment issue records rendered from exported fields. This figure is an evidence visualization, not a direct screenshot of the Jira UI.

JQL purpose	Query	Count
All ACS project issues	project = ACS	46
Product backlog stories	project = ACS AND issuetype = Story	15
Active sprint scope	project = ACS AND Sprint in openSprints()	4
Active sprint incomplete	project = ACS AND Sprint in openSprints() AND statusCategory != Done	4
Active sprint completed	project = ACS AND Sprint in openSprints() AND statusCategory = Done	0
Unresolved bugs	project = ACS AND issuetype = Bug AND statusCategory != Done	0
Release 1.0 label group	project = ACS AND labels = "release-1.0"	4
Release 1.1 label group	project = ACS AND labels = "release-1.1"	4

JQL purpose	Query	Count
Release 1.2 label group	project = ACS AND labels = "release-1.2"	7
Experiment 8 label	project = ACS AND labels = "experiment-8"	1
Experiment 9 label	project = ACS AND labels = "experiment-9"	1
Experiment 10 label	project = ACS AND labels = "experiment-10"	1

A.1 Jira workflow counts

Status	Count	Meaning
Done	19	Jira issue status Done, across experiment records, subtasks and backlog issues.
In Progress	6	Work is actively tracked; includes ACS-11, ACS-20, ACS-21, two Sprint 2 Stories and portfolio Epic ACS-28.
To Do	21	Unstarted backlog work or external verification subtasks.
Total	46	All issues returned by project = ACS in the export query.

A.2 Sprint export

Field	Sprint 1 (ID 1)	Sprint 2 (ID 2)
Name	Sprint 1 - Accounts	Sprint 2 — Alumnex Core
State	Closed	Active
Start	13 Oct 2026, 09:00 IST	13 Oct 2026, 09:30 IST
End planned	20 Oct 2026, 18:00 IST	20 Oct 2026, 18:00 IST
Completion timestamp	13 Oct 2026, 09:13 IST (Jira recorded completeDate)	Not complete
Planned points	8	16
Result	0/2 Stories accepted	4 assigned Stories; 2 In Progress, 2 To Do, 0 Done

A.3 Release version export

Version ID	Version name	Release label	Stories	Points	Released
10037	Release 1.0 — Identity and Profile Foundations	release-1.0	4	21	No
10038	Release 1.1 — Mentorship and Career Enablement	release-1.1	4	16	No
10039	Release 1.2 — Developer Intelligence and Community	release-1.2	7	34	No

Appendix B — Complete Jira Issue Export

The CSV/JSON files linked with this report provide this same structured issue inventory. Fields are: issue key, issue type, summary, status, priority, assignee display name, parent key, Story point estimate where applicable, created/updated timestamps and browse URL. A blank field means the particular export read did not return a value or the field is not applicable; it is not filled by assumption.

Key	Type	Summary	Status	Priority	Assignee	Parent	Pts
ACS-1	Epic	Identity, Profiles and Governance	To Do	High	subhadharanis.24cse	—	—
ACS-2	Epic	Mentorship and Career Opportunities	To Do	High	SRIMATHI D 24CSR302	—	—
ACS-3	Epic	Developer Intelligence and Community Engagement	To Do	High	THIRUMURUGAN A 24CSR330	—	—
ACS-4	Task	Experiment 1 — Product Backlog and Story Refinement	Done	High	SRIMAN VEL	ACS-28	—
ACS-5	Task	Experiment 2 — Release Plan	Done	High	SRIMATHI D 24CSR302	ACS-28	—
ACS-6	Task	Experiment 3 — Sprint Planning for Alumnex Connect	Done	High	SRIMATHI D 24CSR302	ACS-28	—
ACS-7	Task	Experiment 4 — Sprint Execution and Workload Management	Done	High	THIRUMURUGAN A 24CSR330	ACS-28	—
ACS-8	Task	Experiment 5 — JQL Queries	Done	High	THIRUMURUGAN A 24CSR330	ACS-28	—
ACS-9	Task	Experiment 6 — Sprint Review	Done	High	SRIMAN VEL	ACS-28	—
ACS-10	Task	Experiment 7 — Sprint Closure and Next Sprint	Done	High	subhadharanis.24cse	ACS-28	—
ACS-11	Task	Experiment 8 — Alumnex Connect Business Use-Case Model	In Progress	High	subhadharanis.24cse	ACS-28	—
ACS-12	Story	User Registration and Email Verification	In Progress	Medium	SRIMAN VEL	ACS-1	5
ACS-13	Story	Secure Login, OAuth and Session Lifecycle	In Progress	Medium	SRIMAN VEL	ACS-1	3
ACS-14	Story	Mentorship Requests, Matching and Reviews	To Do	Medium	SRIMAN VEL	ACS-2	5
ACS-15	Story	Career Opportunities and Application Tracking	To Do	Medium	SRIMAN VEL	ACS-2	3
ACS-16	Story	DevPulse, Developer Profiles and Activity Goals	To Do	Medium	subhadharanis.24cse	ACS-3	5
ACS-17	Story	AI Career Support and Resume Analysis	To Do	Medium	THIRUMURUGAN A 24CSR330	ACS-2	5

Key	Type	Summary	Status	Priority	Assignee	Parent	Pts
ACS-18	Story	Real-Time Messaging and Notification Preferences	To Do	Medium	THIRUMURUGAN A 24CSR330	ACS-3	3
ACS-19	Story	Events and Google Calendar Synchronization	To Do	Medium	SRIMAN VEL	ACS-3	3
ACS-20	Task	Experiment 9 — Alumnex Connect Conceptual Classes and UML Domain Model	In Progress	High	subhadharanis.24cse	ACS-28	—
ACS-21	Task	Experiment 10 — Alumnex Source Study, Python/MySQL Lab and Testing	In Progress	High	THIRUMURUGAN A 24CSR330	ACS-28	—
ACS-22	Subtask	Registration — Define validation and duplicate checks	Done	Medium	SRIMAN VEL	ACS-21	—
ACS-23	Subtask	Registration — Implement and document endpoint contract	Done	Medium	SRIMAN VEL	ACS-21	—
ACS-24	Subtask	Registration — Unit and integration test plan	Done	Medium	SRIMAN VEL	ACS-21	—
ACS-25	Subtask	Login — Credential verification	Done	Medium	SRIMAN VEL	ACS-21	—
ACS-26	Subtask	Login — Session/token behavior	To Do	Medium	SRIMAN VEL	ACS-13	—
ACS-27	Subtask	Login — Authentication test cases	Done	Medium	SRIMAN VEL	ACS-21	—
ACS-28	Epic	Alumnex Connect — Scrum Lab Experiments 1–10 Portfolio	In Progress	High	SRIMAN VEL	—	—
ACS-29	Subtask	JQL — Sprint Scope and Status Queries	Done	High	THIRUMURUGAN A 24CSR330	ACS-8	—
ACS-30	Subtask	JQL — Release and Experiment Queries	Done	Medium	subhadharanis.24cse	ACS-8	—
ACS-31	Subtask	JQL — Backlog and Defect Queries	Done	Medium	SRIMATHI D 24CSR302	ACS-8	—
ACS-32	Subtask	Sprint 1 Retrospective and Improvements	Done	Medium	THIRUMURUGAN A 24CSR330	ACS-10	—
ACS-33	Subtask	Definition of Ready and Definition of Done	Done	High	SRIMATHI D 24CSR302	ACS-10	—
ACS-34	Subtask	Experiment 10 — Offline Test Evidence	Done	High	SRIMAN VEL	ACS-21	—
ACS-35	Subtask	Experiment 10 — Run Against Real MySQL	To Do	High	subhadharanis.24cse	ACS-21	—
ACS-36	Subtask	Sprint Review — Agenda and Date	Done	High	SRIMATHI D 24CSR302	ACS-9	—
ACS-37	Story	Profiles, Connections and Role-Aware Access	To Do	High	SRIMAN VEL	ACS-1	5
ACS-38	Story	Admin Governance, Moderation and Analytics	To Do	High	subhadharanis.24cse	ACS-1	8

Key	Type	Summary	Status	Priority	Assignee	Parent	Pts
ACS-39	Story	Forum and Tech Hub Knowledge Sharing	To Do	Medium	SRIMATHI D 24CSR302	ACS-3	5
ACS-40	Story	Project Showcase and Collaboration	To Do	Medium	srimanlumar06	ACS-3	5
ACS-41	Story	Competitive Contests and Leaderboards	To Do	Medium	SRIMATHI D 24CSR302	ACS-3	5
ACS-42	Story	Capacitor Mobile App and Native Push Delivery	To Do	High	SRIMAN VEL	ACS-3	8
ACS-43	Story	Institution and Business Directory	To Do	Medium	subhadharanis.24cse	ACS-2	3
ACS-44	Subtask	Experiment 5 — Save Native JQL Filters in Jira	To Do	High	—	ACS-8	—
ACS-45	Subtask	Experiment 6 — Run Live Alumnex Demo Smoke Test	To Do	High	—	ACS-9	—
ACS-46	Subtask	Experiments 8–9 — Attach UML Diagram Exports to Jira	To Do	High	—	ACS-20	—

Appendix C — Direct Jira and Source Links

Jira key	Issue / evidence record	Status
ACS-28	Alumnex Connect — Scrum Lab Experiments 1–10 Portfolio	In Progress
ACS-4	Experiment 1 — Product Backlog and Story Refinement	Done
ACS-5	Experiment 2 — Release Plan	Done
ACS-6	Experiment 3 — Sprint Planning for Alumnex Connect	Done
ACS-7	Experiment 4 — Sprint Execution and Workload Management	Done
ACS-8	Experiment 5 — JQL Queries	Done
ACS-9	Experiment 6 — Sprint Review	Done
ACS-10	Experiment 7 — Sprint Closure and Next Sprint	Done
ACS-11	Experiment 8 — Alumnex Connect Business Use-Case Model	In Progress
ACS-20	Experiment 9 — Alumnex Connect Conceptual Classes and UML Domain Model	In Progress
ACS-21	Experiment 10 — Alumnex Source Study, Python/MySQL Lab and Testing	In Progress
ACS-35	Experiment 10 — Run Against Real MySQL	To Do
ACS-36	Sprint Review — Agenda and Date	Done
ACS-44	Experiment 5 — Save Native JQL Filters in Jira	To Do
ACS-45	Experiment 6 — Run Live Alumnex Demo Smoke Test	To Do
ACS-46	Experiments 8–9 — Attach UML Diagram Exports to Jira	To Do

C.1 Repository files reviewed

- Repository: https://github.com/SrimanKumarV/FSD/tree/master/alumni-connect-website-main
- README: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/README.md
- Alumnex project report: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/Alumnex_Connect_Project_Report.md
- Backend server: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/backend/server.js
- Auth routes: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/backend/routes/auth.js
- Mentorship routes: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/backend/routes/mentorship.js
- Jobs routes: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/backend/routes/jobs.js
- Activity routes: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/backend/routes/activity.js

- Google Calendar routes: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/backend/routes/googleCalendar.js
- GitHub CI workflow: https://github.com/SrimanKumarV/FSD/blob/master/alumni-connect-website-main/.github/workflows/test.yml
- Live demo: https://alumnex-connect.onrender.com
- Backend health route: https://alumnex-backend-9y5t.onrender.com/api/health

C.2 Companion exports and evidence files

Included separately with the report:

- ACS_Jira_Export_2026-10-14.csv — issue data extract for spreadsheet review.
- ACS_Jira_Export_2026-10-14.json — structured snapshot of issues, JQL result counts, sprint records and version records.
- ACS_Scrum_Experiments_Artifacts.zip — source package for the Python/MySQL lab, test result log, Alumnex-specific use-case/domain diagram exports and diagram sources.